

SimForge: A reproducible, cross-simulator benchmarking framework for urban traffic simulation

Phanidhar Akula ¹✉

¹ Department of Computer Science and Software Engineering, Miami University, Oxford, OH, USA ✉
Corresponding author

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Summary

SimForge is a Python framework for reproducible, cross-simulator benchmarking of urban traffic simulators. From a single canonical description of a city, a five-file bundle (road network, travel demand, traffic signals, run configuration, and a SHA-256 hash manifest) generated from open data such as OpenStreetMap, the U.S. Census, and LandScan population grids, SimForge drives three structurally different simulators, SUMO ([Krajewicz et al., 2012](#)), MATSim ([Horni et al., 2016](#)), and DTALite ([Zhou & Taylor, 2014](#)) (via Path4GMNS ([Lu & Path4GMNS Contributors, 2024](#))), from one shared, validated input. The fairness-critical machinery lives in a shared layer rather than in the engines: a strongly-connected-component and feasibility filter produces a single admissible trip set that every engine must run, and shared routes are computed once with a deterministic breadth-first search and reused across engines, so the per-engine adapters only translate the already-fair, already-routed inputs into each simulator's native format. Before any comparison is reported, a machine-checked fairness audit (questions Q1 through Q5) verifies that every engine received byte-identical inputs and computed a byte-identical feasible trip set. Each benchmark cell runs inside a content-addressed container so that results are reproducible by construction, and SimForge reports cross-engine fidelity using the root-mean-square error, the GEH statistic ([Dowling et al., 2004](#)), and the Kolmogorov-Smirnov distance ([Smirnov, 1948](#)), alongside scalability, throughput, and a per-cell reproducibility index computed over repeated seeds with confidence intervals, all collected into an automatically emitted scorecard.

Statement of need

Traffic simulators each encode decades of modeling expertise, yet they disagree, and a reported difference between two of them is hard to interpret. When two engines report travel times that differ by, for example, 10% on the same scenario, that difference conflates the one quantity a researcher wants to isolate, a genuine paradigm-level disagreement between the engines, with several confounds: each engine may silently read the inputs differently, ship different default settings, exhibit floating-point or thread-scheduling non-determinism, or drift between undocumented software versions. Existing cross-simulator studies, including single-engine calibration efforts, small two-engine comparisons, and feature surveys ([Bazzan & Klügl, 2014](#)), generally do not programmatically verify that the engines received identical inputs, so a measured difference cannot be confidently attributed to the modeling paradigm.

SimForge closes this attribution gap with four mechanisms: a canonical input schema with field-level validators, deterministic fixed-seed adapters, a machine-checked Q1 to Q5 fairness audit executed before every comparison, and a content-addressed container that pins code, toolchain, and data by digest. Together these make a cross-simulator difference attributable to the modeling paradigm rather than to formatting, tuning, randomness, or version drift, in the spirit of community benchmarks such as MLPerf ([Mattson et al., 2020](#)) and the FAIR

principles for reproducible research (Pineau et al., 2021; Wilkinson et al., 2016). SimForge also generates structured, activity-based travel demand (home-to-work, home-to-school, and chained trips) from open population and land-use data using a CITYSCAPE-style model (Rao, 2023), so that demand is realistic and reproducible rather than uniformly random.

The framework targets transportation researchers who need defensible cross-engine comparisons and practitioners who must choose a simulator for a specific planning decision. SimForge has been exercised across three cities and five demand tiers spanning three orders of magnitude (1,000 to 500,000 trips), and the substantive research findings from those experiments are reported separately. The point relevant here is the capability the software provides: because every cross-engine comparison rests on a passing fairness audit and a pinned, content-addressed container, any difference it surfaces is reproducible and attributable to the modeling paradigm by construction, rather than to formatting, tuning, randomness, or version drift.

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